



Tecnológico de Monterrey

Reporte:
Actividad 3: ACL Grupal

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Instrucciones:

Dentro de la configuración inicial del laboratorio, habrá una mesa que funcione como "internet". Esta tendrá un router con una interfaz abierta con la ip 10.100.0.100/24 y tendrá también una interfaz a la red 10.200.0.0/24, con un servidor en la computadora 10.200.0.2. La interfaz a la 10.100.0.0/24 tendrá un switch donde se conectarán las demás computadoras.

1. Coloca un router y un switch en tu mesa. Tu router debe tener una interfaz al switch central y otra al switch en tu mesa. Para la interfaz al switch central, pon la ip 10.100.0.X/24, donde X será tu número de mesa. Para la interfaz interior debes tener la red 10.X.0.0/24.
2. Agrega las access list necesarias para lograr lo siguiente:
 - a. Bloquear pings a la red 10.200.0.0/24
 - b. Permitir pings desde cualquier red. La computadora conectada a la red 10.X.0.0 debe poder contestar ping.
 - c. Permitir la salida web (http) hacia la computadora 10.200.0.2.
 - d. Bloquear cualquier otro tráfico

Configuración de dispositivo final:

Red: 10.3.0.0

Gateway: 10.3.0.1

Dispositivo final: 10.3.0.2 /24

IP assignment:	Manual	
IPv4 address:	10.3.0.2	Edit
IPv4 mask:	255.255.255.0	
IPv4 gateway:	10.3.0.1	
DNS server assignment:	Manual	
IPv4 DNS servers:	0.0.0.0 (Unencrypted)	Edit

Ping al dispositivo final de la red central (antes de ser bloqueada):

(Desde la 10.3.0.2 a la 10.200.0.2)

```
C:\Users\pedro>ping 10.200.0.2

Pinging 10.200.0.2 with 32 bytes of data:
Reply from 10.200.0.2: bytes=32 time=3ms TTL=62
Reply from 10.200.0.2: bytes=32 time=1ms TTL=62
Reply from 10.200.0.2: bytes=32 time=3ms TTL=62
Reply from 10.200.0.2: bytes=32 time=1ms TTL=62

Ping statistics for 10.200.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

Ping al gateway del dispositivo final de la red central (después de ser bloqueada):

(Desde la 10.3.0.2 a la 10.200.0.1)

```
C:\Users\pedro>ping 10.200.0.1

Pinging 10.200.0.1 with 32 bytes of data:
Reply from 10.3.0.1: Destination net unreachable.
Reply from 10.3.0.1: Destination net unreachable.
Reply from 10.3.0.1: Destination net unreachable.
Reply from 10.3.0.1: Destination net unreachable.
```

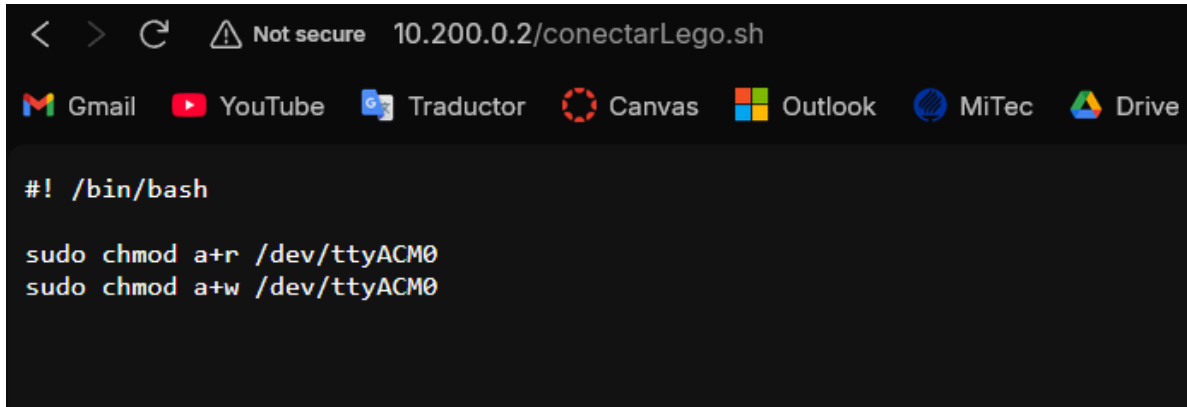
Ping a un dispositivo final de otra red: (Desde la 10.3.0.2 a la 10.2.0.2)

```
C:\Users\pedro>ping 10.2.0.2

Pinging 10.2.0.2 with 32 bytes of data:
Reply from 10.2.0.2: bytes=32 time=5ms TTL=61
Reply from 10.2.0.2: bytes=32 time=3ms TTL=61
Reply from 10.2.0.2: bytes=32 time=3ms TTL=61
Reply from 10.2.0.2: bytes=32 time=3ms TTL=61

Ping statistics for 10.2.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 3ms
```

Acceso al servicio HTTP tras realizar la access list:



```
< > ↻ ⚠ Not secure 10.200.0.2/conectarLego.sh

Gmail YouTube Traductor Canvas Outlook MiTec Drive

#! /bin/bash

sudo chmod a+r /dev/ttyACM0
sudo chmod a+w /dev/ttyACM0
```

Configuración del router de la red del equipo (10.3.0.0):

```
None
Router# show run
Building configuration...

Current configuration : 6266 bytes
!
! Last configuration change at 02:58:14 UTC Sat Apr 18 2026
!
version 17.6
service timestamps debug datetime msec
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname Router
!
boot-start-marker
```

```

boot-end-marker
!
no aaa new-model
!
login on-success log
!
subscriber templating
vtp version 1
multilink bundle-name authenticated
!
crypto pki trustpoint TP-self-signed-2270750389
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2270750389
  revocation-check none
  rsakeypair TP-self-signed-2270750389
!
crypto pki trustpoint SLA-TrustPoint
  enrollment pkcs12
  revocation-check crl
!
crypto pki certificate chain TP-self-signed-2270750389
  certificate self-signed 01
    30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
    31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
    69666963 6174652D 32323730 37353033 3839301E 170D3236 30343138 30323237
    32355A17 0D333630 34313730 32323732 355A3031 312F302D 06035504 03132649
    4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 32373037
    35303338 39308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
    0A028201 0100990B D2E88170 751B3C6A 25903259 A2BC043D 906A4BC6 41C5657A
    5CE3A3A7 CCB22B54 FD6D5A85 091C794B BA93D72C 4798E70D E261CEF3 15B20BA4
    B912A9CB 532B286E 5D5DE8A7 8165289E 57580A0A 9960345A C354110A FEEF05FC
    7763B346 FEFD1B3E E9102DA7 3F96CE78 15BEFC33 936CE4EF E6479C74 706B7BC8
    A91C59A1 D8C3D4F7 12360311 6DD4805B 8BF0D6AC DD7D4C69 07701243 91BF432C
    97233109 A5FB7EF9 A726D582 F6E74EFB 283DF5FC 3DD5AA5C 8B03CAAA D72F63EF
    4E13273A 4C0F394B 36EF94E5 2E9304CB 4B14CDE6 AEAA1325 5635FD04 E8956127
    F6A9C4F9 261319A9 E4ABF24C 168CF175 8E045297 2FD09B3E C07DFFAF 61CD1D97
    27EC539C 009B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
    301F0603 551D2304 18301680 14A645CC 2814B545 5219D894 A22E309A C5FF8309
    78301D06 03551D0E 04160414 A645CC28 14B54552 19D894A2 2E309AC5 FF830978
    300D0609 2A864886 F70D0101 05050003 82010100 05EB5F0E 48C81B88 DB66CE50
    62D0233F 6077ED50 D756B9A1 E09DF3C3 005F999D 811019A0 978D649B 0439322C
    7E437BC0 BC547E2B F9540541 96012E5D 6B1D87CF C6442DF9 37178E19 0CFACCCA
    0A386AD2 274EC76C 1BE38BBC D8334892 B0C3D2C5 1776DA58 E489F43B D7BA40CE
    91DD739D DD001FC7 93230B3A EE14A6CA 20C5038B DBE67FFC 75358F8F 634C97BA
    1D8AA634 61D24888 FA44A6DC 086C01EE 891FBAF6 66704F4B FB11C29B 21D5BD9D
    93C062D5 1EA7B14E 8DBD011B 36607709 C9CC2DE0 6591CEE8 CB5FA94B 833B4583
    93CC3471 A544DF6F 336B7000 C30CD2C9 1506B959 4F1F7F8E C8AE519B 59EB593D
    F01F51F2 80FFBD9B E0291B56 DAEA281C B6A15ABB
  quit
crypto pki certificate chain SLA-TrustPoint
  certificate ca 01
    30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
    32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
    6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934

```

```
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBA3 700A8BF7 D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECAC02 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEB7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28
quit
!
license udi pid C8200L-1N-4T sn FJC282212SP
memory free low-watermark processor 67708
!
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
redundancy
mode none
!
interface GigabitEthernet0/0/0
ip address 10.100.0.3 255.255.255.0
negotiation auto
!
interface GigabitEthernet0/0/1
ip address 10.3.0.1 255.255.255.0
ip access-group 101 in
negotiation auto
!
ip http server
ip http authentication local
ip http secure-server
ip forward-protocol nd
ip route 0.0.0.0 0.0.0.0 10.100.0.100
!
ip access-list extended 101
10 deny icmp 10.3.0.0 0.0.0.255 10.200.0.0 0.0.0.255
20 permit icmp 10.3.0.0 0.0.0.255 any
30 permit tcp 10.3.0.0 0.0.0.255 host 10.200.0.2
```

```

40 permit udp 10.3.0.0 0.0.0.255 host 10.200.0.2
!
control-plane
!
line con 0
  logging synchronous
  stopbits 1
line aux 0
line vty 0 4
  login
  transport input ssh
line vty 5 14
  login
  transport input ssh
!
call-home
  ! If contact email address in call-home is configured as sch-smart-licensing@cisco.com
  ! the email address configured in Cisco Smart License Portal will be used as contact
  email address to send SCH notifications.
  contact-email-addr sch-smart-licensing@cisco.com
  profile "CiscoTAC-1"
  active
  destination transport-method http
!
end

```

Ip routes configuradas:

```

None
Router# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.100.0.100 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.100.0.100
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C      10.3.0.0/24 is directly connected, GigabitEthernet0/0/1
L      10.3.0.1/32 is directly connected, GigabitEthernet0/0/1
C      10.100.0.0/24 is directly connected, GigabitEthernet0/0/0
L      10.100.0.3/32 is directly connected, GigabitEthernet0/0/0

```